

Antonio Ferrão Neto

THE METAPHOR OF THE BUBBLELAND

A PROTOSCIENTIFIC APPROACH

The Argument

Volume I

 **ImpNet**

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A Protoscientific Approach

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*I dedicate this work
to GOD,*

*to my parents LEÔNIDAS FERRÃO (in memoriam) and
RUTH TRIGO FERRÃO (in memoriam),*

to my siblings SÔNIA, LEÔNIDAS and SANDRA,

to my daughter KETELLY,

*and to all who believe that
knowledge and imagination
can transform the world.*

About the cover

The cover image depicts the lake of the **Metaphor of the Bubbleland**, the thread that runs through the entire work. It is on the surface of that still water — where bubbles link to one another, forming and dissolving clusters that move across the water's surface, and waves spread out — that the book finds its central analogy: bubbles as atoms, clusters as bodies, waves as light, time as the mere rhythm of that movement. From the simple act of looking at a lake springs everything that follows.

This cover image was composed with the aid of artificial intelligence. The final artwork is released under a **Creative Commons Attribution 4.0** license (CC BY 4.0): it may be reproduced, adapted and used freely, including commercially, provided the source is credited — *The Metaphor of the Bubbleland*, by Antonio Ferrão Neto.



Note to the reader (*Disclaimer*)

Although it bears **the characteristics of scientific writing**, this work is essentially **protoscientific**; that is, it is built upon exploratory hypotheses. The ideas presented here should not be interpreted as established theories, but rather as **speculative hypotheses** in the process of being formulated, grounded in observations, analogies, and inferences that await experimental validation.

The purpose is to **stimulate reflection and debate** about the limits of current knowledge and the possibilities of new theoretical approaches. This text is not a substitute for peer-reviewed scientific publications, nor should it be understood as technical or professional guidance.

All of the conceptual content, including metaphorical models, interpretations, and propositions, constitutes the author's **own original work**, even where it draws on external references used solely as bibliographic support.

Should any proposition presented here come to be scientifically confirmed in the future, such confirmation would not alter the fact that, at the time of this publication, it remains a rigorous speculative exercise. Any coincidence between the hypotheses described here and future discoveries does not imply their retroactive scientific validation as of the date of this work.



Preface

Therefore, I do not believe I have done a useless work in writing this Memoir; I only regret that it is too long. However, when I tried to restrain myself, I fell into obscurity; I preferred to be considered a little verbose.

— H. POINCARÉ (POINCARÉ, 1895)

As Poincaré pointed out in his article *Analysis Situs*, it is not always possible to write a concise text without losing clarity. Works involving a high level of abstraction require readers to confront paradigm shifts, observe facts from new perspectives, and question concepts previously established by the scientific community.

For Isaac Newton, alchemy was not naive mysticism, but a legitimate attempt to understand the intimate structure of matter, its transformations, and its hidden forces.¹ Although this work is classified as protoscientific, this designation should not be confused with an appeal to mysticism, esotericism, metaphysics, or supernat-

¹ Historiographical studies indicate that more than one million words of Newton's manuscripts are devoted to alchemy, a volume far greater than his published output in mechanics, optics, and mathematics. Newton copied, commented on, translated, and reinterpreted numerous medieval and Renaissance alchemical treatises, treating alchemy as an integral part of the natural philosophy of his time. These writings remained unpublished for centuries, in part due to the risks of intellectual and religious persecution associated with such investigations. For detailed analyses, see Westfall (WESTFALL, 1980), Dobbs (DOBBS, 1975), and the collection of manuscripts assembled by Keynes, now preserved in the Cambridge University Library (KEYNES, 1936).

ural explanations — themes deliberately absent from this book. The term is employed here in its historical and epistemological sense: that of an effort situated on the frontier between what is already formalizable and what is not yet fully testable. In this respect, the work is closer to the tradition of natural philosophy than to consolidated scientific theories. A classic example is Isaac Newton, whose contributions founded Classical Mechanics and Calculus, but who also worked extensively in domains today considered pre-scientific, formulating laws of enormous predictive power before the existence of a complete causal foundation. Analogously, this book proposes conceptual structures, interpretive models, and hypotheses coherent with current scientific knowledge, explicitly acknowledging their empirical limitations and their exploratory character, as part of a legitimate process of theoretical anticipation.

The importance of early publication

Beware the barrenness of a busy life.

— SOCRATES (attribution not confirmed)

Donald Knuth, one of the greatest specialists in Computer Science, began in 1968 his monumental work *The Art of Computer Programming* (KNUTH, 2023), planned for seven volumes; by 2026, only four had been published. Had he waited for the complete conclusion before publishing, humanity might have been deprived of this knowledge for more than half a century.

The Metaphor of the Bubbleland allows hypotheses to be formulated through speculation, a fundamental stage in the scientific process. Even if some proposals seem, at first glance, radical or bold, the history of science shows that such initiatives are vital to the advancement of knowledge. However, without experimental validation, speculations cannot be considered science. Despite this, science has a history of early publication of revolutionary ideas:

- **Leonardo da Vinci:** Between the 15th and 16th centuries, he recorded in his notebooks hundreds of machines, mechanisms, and physical principles extremely advanced for the time — including flying devices, automata, war machines, and hydraulic systems. Most of these ideas remained merely unpublished manuscripts for centuries, functioning as a form of “early publication” of concepts that would only become understood or technically feasible much later.² (RETI, 1974; KEMP, 1981; LAURENZA; TADDEI; ZANON, 2006).

² His notebooks, although not published in the modern sense, circulated among disciples and

- **Galileo Galilei:** He published his observations in *Sidereus Nuncius* (1610) before completely confirming the discoveries about the satellites of Jupiter and the phases of Venus.³ (GALILEI, 1610).
- **Amedeo Avogadro:** In 1811 (AVOGADRO, 1811), he formulated the hypothesis that equal volumes of gases, under the same conditions of temperature and pressure, contain the same number of molecules — an idea extremely premature for the time, ignored for nearly half a century and only validated after the Karlsruhe Congress (1860) (CANNIZZARO, 1858).
- **Charles Darwin and Alfred Russel Wallace:** They presented the theory of evolution by natural selection to the Linnean Society of London before the collection of all empirical evidence (DARWIN; WALLACE, 1858).
- **Albert Einstein:** He published the theory of special relativity before its experimental confirmation (EINSTEIN, 1905b).
- **Eric Drexler:** In *Nanosystems: Molecular Machinery, Manufacturing, and Computation* (1992), he presented visionary concepts about molecular nanotechnology, many of which are still under development (DREXLER, 1992).

Early publications can boost science by:

- Sharing knowledge and stimulating debates;
- Accelerating experiments and validation;
- Ensuring priority over discoveries;
- Facilitating funding and collaborations.

Just as Leonardo da Vinci, Knuth, Galileo, Avogadro, Darwin, Einstein, Drexler, and many other scientists did, this work is published early to foster discussion and development. Scientific validation will be essential to consolidate its hypotheses — as in every genuinely scientific process. Pioneering ideas, such as the existence of subatomic particles or the structure of DNA, were initially received with skepticism until rigorous experiments transformed them into pillars of knowledge. Likewise, the hypotheses presented here will only leave the protoscientific stage when tested by replicable, measurable methods that are coherent with the internal logic of science. If revolutionary ideas are set aside because of overly busy lives, we will lose

patrons and represented a repository of ideas that anticipated technologies that would only be realized centuries later.

³ These observations were made with the telescope that Galileo had built, allowing him to see details of the sky never before observed. Although his observations were published, he continued investigating to consolidate them. This reflects the essence of the scientific method, which involves constant review and validation.

opportunities for advancement — as has happened so many times. Gregor Mendel published his work on heredity in 1866, but it remained ignored for decades. Ignaz Semmelweis proposed the disinfection of hands before childbirth, but his warning was disregarded, at the cost of thousands of lives. History shows that being right is not enough: someone must have the time — and the courage — to listen.

Nowadays, formulating good questions is more important than knowing how to answer them. The answers, almost always, can be obtained quickly through AIs or internet search tools, while the creation of original and profound questions is a far more demanding task.⁴ As this is a protoscientific book, this volume functions as a **“manual of questions”**. Here, there are no wrong questions: each question is a legitimate disturbance on the surface of the unknown, an indispensable step in the process of understanding what we do not yet know.

The contemporary scientific method, widely standardized throughout the world, brought reproducibility and rigor, but also narrowed the paths through which new ideas can emerge. As Feyerabend (1975) warned, excessive standardization turns scientific investigation into a methodological monoculture, little receptive to radical or intuitive approaches. This is why the initial phase of ideas — protoscience — usually occurs in silence: researchers avoid disclosing still fragile conjectures, fearing the loss of priority or recognition.

It is at this point that *The Metaphor of the Bubbleland* offers a profound interpretation. Protoscience corresponds to the stage in which each researcher initially works in a *formulation space* of their own — a territory still in formation, marked by uncertainties, constant revisions, and ideas that do not yet possess a clear or communicable form. At this stage, hypotheses arise as small local ripples, fragile and without external validation. Only when they begin to acquire greater clarity and consistency can these formulation spaces draw closer together, allowing the dialogue, criticism, and collective construction of knowledge that characterizes mature science.

This book deliberately assumes the point of view of that unstable region of Bubbleland. It does not offer consolidated answers, but cultivates the ripples that precede any real discovery. It is not science proper — but all science begins precisely like this.

⁴ An exception to this rule occurs when the answer must be absolutely original — that is, when there does not exist, anywhere in the world, a prior formulation that resolves the proposed question, and the person interrogated is able to produce a genuinely unprecedented answer. This scenario is possible, but extremely rare.

Note on the *sui generis* character of this book

As far as the author has been able to verify, there is no editorial precedent that combines, in a single work, *all* of the following elements:

1. **Progressive pedagogical metaphors with explicit deconstruction** — Bubbleland is not a decorative device, but a cognitive structure with a deliberate life cycle: construction, use, and formal dismantling (Chapter 6).
2. **Neologisms with formal definition** — terms such as *nomia* are introduced with rigorous mathematical definition ($S, \mathcal{D}, C$), not as metaphors.
3. **Structured ontological speculation** — the hypotheses about parallel realities, biomolecule tunneling, and temporal disconnection are not presented as fiction, but as propositions with traceable formal consequences.
4. **Computational proofs of concept embedded in the text** — Python source code, Monte Carlo simulations, and statistical analysis of results serve as argument, not as illustrative appendix.
5. **Scientific-paper formalism within accessible prose** — definitions, theorems, proofs, and $\LaTeX$ notation coexist with narrative language and intuitive examples.

The coexistence is not accidental: the reason an appendix of numerical engineering shares the book with chapters of philosophy is laid out in Section 18.1.

Existing works come close in isolated aspects. *Flatland*, by Edwin Abbott (1884), uses a geometric metaphor for higher dimensions, but without formalism. *Gödel, Escher, Bach*, by Douglas Hofstadter (HOFSTADTER, 1979), combines metaphors, dialogues, and formalism, but without a computational-experimental component. *The Fabric of Reality*, by David Deutsch (DEUTSCH, 1997), defends the Many-Worlds interpretation with philosophical argumentation, but without its own conceptual tools or simulations. *Our Mathematical Universe*, by Max Tegmark (TEGMARK, 2014), classifies levels of multiverse, but is the dissemination of results already published in journals.

None of these ingredients is unprecedented in itself. What appears to be unprecedented is the **combination**: ontological speculation formalized, validated by computational simulation, structured by progressive deconstructible metaphors, within a text that is simultaneously narrative and formally rigorous. This observation does not intend to claim superiority, but only to **inform the reader that there does not exist, as far as is known, a prior editorial model by which to judge this work**. The reader who tries to frame it as “science popularization” will be puzzled by the formalism; as an “academic paper”, will be puzzled by the metaphors; as “science fiction”, will be puzzled by the Monte Carlo simulations. The correct reading key is to accept that this is a *sui generis* work — a hybrid genre, constructed out of necessity, not out of caprice.

As discussed in Section A.16 (Vol. II), the set of ideas developed here proved to be extensive and interdependent, which made its publication in article format unfeasible.

On the absence of critical reviewers

This work was entirely developed by the author, without the support of critical readers or specialized reviewers. For this reason, the observations and suggestions of the readers of this first edition will be analyzed with attention, and those that contribute to the improvement of the hypotheses or to expository clarity may be incorporated in future revisions.⁵

A compendium of hypotheses

This text constitutes a collection of hypotheses driven by *The Metaphor of the Bubbleland*⁶ — a set of propositions to be subsequently verified or refuted. Such hypotheses should not be confused with dogmatic beliefs or unfounded claims. Science progresses not only through experimental confirmation, but also through the uncomfortable questioning of established certainties.

When Galileo pointed his spyglass at the sky, he was not proving anything immediately; he was merely challenging the prevailing model. Likewise, when Heisenberg formulated the Uncertainty Principle, he was calling into question the classical notion of deterministic trajectories. Doubt, when guided by reason and curiosity, is the engine that precedes observation, experimentation, and, often, scientific revolution itself.

However, there is an essential distinction between proposing new approaches and rejecting consolidated knowledge without foundation. Denying evidence and upholding conceptions already refuted by the scientific method does not constitute intellectual boldness, but rather denialism. Thus, although some of the hypotheses explored here may sound radical or even utopian, they remain anchored in the purpose of expanding our understanding of reality — unlike denialism, which limits itself to invalidating without offering coherent alternatives.

⁵ Finding reviewers with training sufficiently broad to encompass the multiple fields gathered here — Advanced Mathematics (especially Group Theory and Computational Mathematics), Electronic Engineering (for the *Method of Successive Zooms*), Philosophy of Science, Epistemology, Quantum Mechanics, Biochemistry, and Molecular Biology, among others — proved to be an extremely difficult task. Furthermore, the controversial and disruptive nature of several themes discussed in this text makes the hesitation of potential collaborators to participate in this review process understandable.

⁶ On hypotheses in the context of *The Metaphor of the Bubbleland*, see Appendix B (Vol. II), in particular Section B.10 (p. 101), as well as the complete catalog of hypotheses in Table 34 (Vol. III) of Appendix AH (p. 391).

This text is protoscientific and, like every protoscientific text, intends to be eventually transformed into science. For this, significant investments and an adequate experimental structure will be necessary.

Some propositions present in this work may challenge traditional models or seem paradoxical from the perspective of conventional science. And, frequently, those who think in an unconventional way are labeled as mad. Many pioneers who transformed Science were initially ridiculed and considered insane.⁷ Someone regarded as mad may simply think differently, “outside the box”, contradicting scientific consensus or, simply, not being fully understood.⁸ After all, it is almost impossible not to seem mad when discussing Quantum Mechanics.

The madness of thinking differently

No great mind has ever existed without a touch of madness.

ARISTOTLE (SENECA, 2005, p. 42)

The relationship between madness and genius has always been an object of reflection. Many scientists and thinkers, in their time, were seen as “mad”, only to have their ideas later validated. The history of science shows that challenging paradigms requires courage and, frequently, leads to ostracism before recognition.

The case of **Galileo Galilei**, whose defense of heliocentrism placed him in conflict with the Church, is a classic example, although today his work is widely accepted (GALILEI, 1610). **Charles Darwin** was ridiculed for proposing that human beings share common ancestors with other primates (DARWIN, 1871; DARWIN, 1859). **Nikola Tesla**, with his futuristic visions, had his contributions treated as extravagant, despite being recognized today as one of the fundamental figures in the development of modern electricity (TESLA, 1919).

The concept of “madness” in science is ambiguous: it can mean an innovative vision that breaks with the established order or, simply, a form of thinking not

⁷ All truth passes through three stages: first, it is ridiculed; second, it is violently opposed; and third, it is accepted as self-evident. — UNKNOWN AUTHOR. This quotation is commonly attributed to Arthur Schopenhauer, but there is no record of it in his writings: the corresponding passage, in the Preface to the first edition of *Die Welt als Wille und Vorstellung* (1818), speaks of **two** long periods of rejection, between which truth is allotted only a brief triumph.

⁸ “And those who were seen dancing were thought to be insane by those who could not hear the music.” — Unknown author. This quotation is commonly attributed to Friedrich Nietzsche, but there are no sources confirming this authorship.

understood by its contemporaries. The difficulty in accepting ideas outside the norm often reflects the cultural and cognitive limits of the era.

Throughout this text, we will discuss how certain hypotheses — especially those that engage with parallel realities, unconventional structures, and new ways of describing the Universe — may initially seem insane. Nevertheless, history suggests that some of the greatest scientific revolutions were born precisely from this place of estrangement.

Scientific “madness”, understood as intellectual boldness, plays an essential role in the advancement of knowledge. Questioning established models, exploring counterintuitive paths, and proposing alternative structures are attitudes that, although they may generate resistance, frequently drive significant discoveries.

Epistemological note on the hypotheses presented

The hypotheses presented throughout this book can be classified, from the perspective of current scientific knowledge, as imaginative speculations still without empirical proof or solid backing in an established body of knowledge. The inclusion of these ideas is intentional and clearly delimited: it is a controlled exercise of speculation, which makes explicit from the outset its provisional and unvalidated character.

These propositions do not intend to replace consolidated theories, nor to compete with the knowledge produced by the scientific community. Their function is complementary: to offer alternative conceptual models that may inspire new questions, suggest possible reorganizations of already known concepts, and point to hypotheses that, in the future, might perhaps be formalized and tested with greater experimental rigor.

From an epistemological standpoint, the text is situated in the domain of proto-science. The ideas presented here should not be interpreted as definitive descriptions of reality, but as heuristic constructions: preliminary conceptual structures, useful for thinking about phenomena that today escape the boundaries of established models. The value of these hypotheses lies, therefore, less in their eventual confirmation and more in their capacity to generate research programs, theoretical debates, and new ways of organizing knowledge.

It is important to emphasize that, at no point does this book propose the abandonment of scientific criteria of validation. On the contrary, it is presupposed that any real advance will depend on rigorous empirical studies, conducted by multidis-

ciplinary teams, with adequate methodology, peer review, and reproducibility of results. Without this path, all the hypotheses discussed here remain on the speculative plane.

Even so, if even a fraction of the propositions developed throughout *The Metaphor of the Bubbleland* comes to find future empirical support, the potential developments could be profound, especially in areas such as molecular biology, neuroscience, fundamental physics, and epistemology.

On the resistance of academia

Although this is a protoscientific work, written by a researcher who operates outside traditional Academia, it is important to recognize the structural limitations of the prevailing academic system in the reception of innovative ideas.

Academia, as an institution dedicated to the production and validation of knowledge, tends to operate with strong inertia. This inertia is not the result of negligence, but a direct consequence of its fundamental commitments: methodological continuity, rigorous validation, reproducibility, and the progressive accumulation of discoveries. This model guarantees solidity to scientific knowledge, but it also imposes natural barriers to the rapid incorporation of proposals that break with established paradigms.

Historically, several ideas today considered fundamental faced initial resistance, both because they challenged prevailing models and because they emerged at moments when the conceptual or technological tools were not yet mature. This delay in acceptance does not stem from ill will, but from an institutional mechanism of protection against errors, exaggerations, or unfounded speculations.

However, recognizing this structural resistance is essential when presenting hypotheses that, by their speculative nature, lie outside the traditional scope of academic research. The propositions developed throughout *The Metaphor of the Bubbleland* do not intend to bypass the scientific model, but to operate in a distinct space: that of protoscience, where preliminary ideas can be formulated, organized, and explored before concrete means exist for their testability.

Thus, academic resistance should not be seen as a hostile obstacle, but as part of the scientific ecosystem. Its role is to ensure methodological solidity, while protoscience, operating in parallel, keeps alive the space of disciplined imagination — the space in which this book is situated.

Note on the use of artificial intelligence in the elaboration of the text

*If I see further, it is by standing on the shoulders of giants —
now multiplied by search engines and artificial intelligence.*

— Adapted from BERNARD OF CHARTRES (12th century) and ISAAC NEWTON (1675/6)

During the writing of this work, the support of artificial intelligence tools was used for purposes of grammatical, orthographic, and stylistic revision, as well as for technical assistance in the organization of the text, research and reading of bibliography, creation of some illustrative images, programming in Python⁹ and preparation of the material in $\text{\LaTeX}$.

It is emphasized, however, that the totality of the ideas, hypotheses, metaphors, interpretations, and conceptual constructions present in the work is of human origin. The role of artificial intelligence was limited to tasks of textual and technical support, functioning as an auxiliary instrument in the systematization and improvement of the presentation of the ideas, and not as a source of original content.

This use dialogues, in a certain way, with the notion of *extended mind*, proposed by Clark and Chalmers (1998), according to which certain external instruments can temporarily integrate the cognitive process by performing functions that, if carried out internally, would be recognized as part of the act of thinking. In this sense, the artificial intelligence used here operates as a functional extension of the author's cognitive capacities — broadening the scope of the research, facilitating the organization of the material, and optimizing mechanical tasks — without, however, replacing the intellectual agency responsible for the propositions developed.

The responsible use of these tools integrates with the spirit of this work, which seeks to explore new ways of thinking, writing, and organizing knowledge. Just like other technologies of intellectual support throughout history — from logarithmic tables and calculators to digital databases — artificial intelligence here plays the role of expanding capacities, without replacing the author or their intellectual responsibility for the content presented.

⁹ One example is Appendix AC (Vol. III), whose programming was entirely generated by an LLM under the author's guidance (see Box 22, p. 234).

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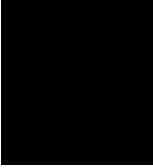
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CHAPTER I



The Metaphor of the Bubbleland

Though this be madness, yet there is method in't.

— WILLIAM SHAKESPEARE, *Hamlet*

*Science is made up of errors, which in turn are the steps
toward truth.*

— JULES VERNE (commonly attributed, no confirmed
primary source)

If the facts don't fit the theory, change the facts.

— ALBERT EINSTEIN (attribution not confirmed)

The fact is dumb. What thinks is thought.

— NELSON RODRIGUES

WE LIVE in an age in which an excess of information is mixed with a scarcity of reflection. The abundance of data, opinions, interpretations, and discourse does not necessarily translate into understanding. On the contrary: the more we surround ourselves with fragmented content and competing narratives, the harder it becomes to build a coherent view of the world.

In this scenario, proposing structured metaphors — such as the Bubbleland Metaphor — is an effort of intellectual organization. It is an attempt to create a conceptual model that lets us examine complex phenomena through new lenses, weaving together science, imagination, and disciplined speculation. The metaphor does not aim to replace scientific theories, but to offer an alternative mental space for thinking about what we still cannot fully describe.

1.1 On hypotheses and boldness

This book presents a series of hypotheses, many of which cannot yet be validated experimentally. Some are working hypotheses; others are speculations grounded in structured analogies, recurring patterns, or plausible interpretations of phenomena still poorly understood. None of them should be read as absolute truth. Throughout the text, the reader will encounter propositions that challenge common sense, models that at first glance seem to contradict known laws of physics, and analogies that connect science, philosophy, and imagination.

The history of science shows, however, that ideas initially deemed improbable, eccentric, or implausible may, in time, prove to be the foundations of new paradigms. Relativity (EINSTEIN, 1905b), Quantum Mechanics (PLANCK, 1901), and the Theory of Evolution (DARWIN, 1859) itself met significant resistance when first presented, whether because they challenged entrenched conceptions or because they demanded a profound restructuring of the scientific outlook.

As is commonly attributed to Eça de Queirós, “what is accepted today as science was yesterday called a chimera.” Regardless of its authorship, the idea it expresses captures a central point: conceptual transformations require intellectual boldness, openness to the new, and a willingness to revisit deep-seated assumptions.

1.2 Metaphors as a tool of discovery

According to Lakoff and Johnson (LAKOFF; JOHNSON, 1980), metaphor is more than a rhetorical device:

For most people, metaphor is a device of the poetic imagination and the rhetorical flourish — a matter of extraordinary rather than ordinary language. Moreover, metaphor is typically viewed as characteristic of language alone, a matter of words rather than thought or action. For this reason, most people think they can get along perfectly well without metaphor. We have found, on the contrary, that metaphor is pervasive in everyday life, not just in language but in thought and action. Our ordinary conceptual system, in terms of which we both think and act, is fundamentally metaphorical in nature (LAKOFF; JOHNSON, 1980, p. 3).

Again, as Lakoff and Johnson (1980, p. 5) puts it:

The essence of metaphor is understanding and experiencing one kind of thing in terms of another.

In short, metaphors:

- are used to describe one thing in terms of another. They may seem absurd, but what they set out to describe is real and not absurd;
- are implicit comparisons;
- allow a deeper, more figurative understanding of the object or concept being described, even when the language used is not literal;
- expand the human capacity for expression and understanding;
- can be imprecise or ambiguous, like any comparison;
- may, at first — before being defined and developed — give rise to doubt or discomfort due to unfamiliarity with their aims;
- should not be interpreted in a literal, denotative, or dictionary sense;
- require a certain level of abstraction to be understood. Some animals considered intelligent can grasp human words literally, but not metaphorically;
- are provisional in nature: they can be discarded once they have played their part, leaving behind only the idea or insight they sought to convey;
- are more than rhetorical devices — they are essential components of the structure of human language.

In this text we will use a few metaphors, the principal one being called the **Bubbleland Metaphor**, initially inspired by another well-known metaphor: **Flatland** (ABBOTT, 1994), often used by physicists to describe what a fourth spatial dimension might be. The *Bubbleland Metaphor*, however, is richer in detail and draws deeper parallels with our own reality. Flatland serves only as an initial inspiration; as we

shall see, the *Bubbleland Metaphor* goes far beyond the mere explanation of additional spatial dimensions.

Flatland has been described as social satire, “science fiction,” and mathematical theory. It is indeed a satire of the Victorian United Kingdom, but it may be more accurate to classify it as a form of fantastic realism rather than science fiction proper. As for its mathematical aspect, its pedagogical value for understanding dimensions beyond the three we experience directly is undeniable.

The *Bubbleland Metaphor*, in turn, has no social-satire aim, though it retains elements of fantastic realism. Its conceptual applications lie much closer to physics than to mathematics. As for understanding the so-called “fourth dimension,” the *Bubbleland Metaphor* inherits the same essential conclusions as Flatland, but dramatically broadens the interpretive scope by connecting multiple possible realities, conceptual interactions, and deeper epistemological structures.

1.3 *The Metaphor of the Bubbleland*

To avoid criticism, do nothing, say nothing, be nothing.

— ELBERT HUBBARD

Let us undertake here a speculative exercise, metaphorical in character, plainly impossible to realize, but whose purpose will become clear at the end, once we accept — if only temporarily — a few absurd premises.

A speculative exercise is not a theory or a practical idea. It is a thought experiment. It may be unfeasible in reality or work only as a metaphor, as is the case here.

Let us imagine an immense lake, as vast as our imagination will allow, with completely still waters — no waves and no wind.

Now suppose there is a surfactant in this water and that many bubbles float on its surface. Naturally, owing to surface tension, some bubbles attract one another, forming clusters. These clusters resemble, in certain respects, real molecules. There are many differences, however: molecules have discrete sizes and types, specific bonds, varied interatomic forces, and an arrangement in three-dimensional space, whereas in this model the bubbles interact in only two dimensions. Even so, these bubbles join together and stay united, as long as they are not disturbed or do not burst.

These bubbles, floating on this virtually infinite lake, form clusters over the surface of the water, leaving empty spaces and spreading across its whole extent. We will call these clusters “objects made of bubbles,” or simply “objects.” The objects have varied shapes, sizes, and compositions, with bubbles of different dimensions.



Figure 1 – A large lake with many bubbles floating on its surface. In the *Bubbleland Metaphor*, the lake would be “virtually infinite,” that is, vastly larger, like an immense and calm ocean.¹



Thus we have two-dimensional objects on the surface of the water, separate or connected in some way. This is the setting of the “two-dimensional bubble world,” which we will call “Bubbleland.” Nothing exists beyond the water and the bubbles. And, for the purposes of the metaphor, these bubbles are extremely stable — they rarely burst.

Now let us push our speculative exercise further: imagine that these bubbles can organize themselves like two-dimensional molecules and form complex structures, without new bubbles being created or destroyed. These structures would be able to reorganize themselves, reusing bubbles discarded from other structures. They could thus feed on air bubbles and even self-replicate, forming a kind of life based not

¹ But since this is a metaphor, at certain points we will consider the lake to be the size of our Universe, that is, billions of light-years across.

on molecules like the ones we know, but on bubbles, which would play the role of atoms.

Let us go even further: suppose that these beings made only of air bubbles are intelligent. We will call these beings “Bubblelanders”².

Once these barriers of plausibility are overcome, we would have beings that inhabit and perceive only two dimensions — as in *Flatland*. All they see are the structures formed by the bubbles. For these beings, it would be extremely difficult to conceive that the lake in which they live belongs to a reality distinct from the one they are able to perceive. They merely “see” clusters of bubbles.

Even if they developed an extremely powerful microscope, capable of revealing the structure of their matter, all they would see would be the bubbles³ — and not the H₂O and surfactant molecules that make up those bubbles and which, in turn, form their bodies and their entire world. They would not naturally grasp the third dimension, nor anything of the “outside world,” which would be the “human world.”

1.3.1 Dimensions beyond the third: the historical resistance

In his book *Hyperspace*, Kaku (1994, p. 30–56) recounts the history of the multiple dimensions of our space, which he calls Hyperspace. He tells of Carl Friedrich Gauss, one of the greatest mathematicians of all time, who asked his student Georg Friedrich Bernhard Riemann to give an oral presentation on the “Foundations of Geometry” as an alternative to Euclidean Geometry. A few decades earlier, Gauss had spoken privately with his colleagues about “bookworms” that would live exclusively in a two-dimensional world. But this generalization of Euclidean Geometry to multiple dimensions was never published by Gauss because, being very conservative, he feared the subject would provoke great outrage among the other mathematicians of his day. This inhibited Gauss, who called those likely opponents “Boeotians” (KAKU, 1994, p. 35).

And, as Kaku (1994, p. 56) comments regarding the book *Flatland*, Gauss’s “bookworms” became “Flatlanders.” The Boeotians, whom Gauss so feared, became the High Priests of *Flatland*, who would persecute and punish anyone who dared mention the invisible third dimension.

It is not yet known who the “Boeotians of Bubbleland” will be. But the golden rules for not becoming a “Boeotian of Bubbleland” are:

² See the Glossary for the meaning of “Bubblelander” and other terms used in this text.

³ They would not be able to observe the bubbles individually because they would be absurdly small, exactly as atoms are to us humans.

1. To be able to abstract one's thinking beyond what one experiences day to day. One cannot claim that everything that exists must necessarily be intuitive to us. The Universe is not as we would like it to be, nor as our intuition would be able to comprehend it. The Universe simply is what it is. It is the filters of experimentation that guide the scientific method;
2. To be able to grasp the non-literal meaning of a metaphor, always bearing in mind that it comparatively describes one thing (the source)⁴ in terms of another (the target): one must develop the thinking within the source while reasoning in terms of the target. In the case of Bubbleland, we will develop the thinking within the source (Bubbleland), which is a flat world, impossible to actually realize, while always reasoning in terms of what it stands for in the target of the metaphor, which is our tangible, three-dimensional reality (setting aside time and other dimensions) — a reality that is possible, since it is where we live.

In the case of the parallel-realities hypothesis, it is necessary to delimit the word “exist” semantically (see Box 1). For instance, one might hold that only tangible objects in our reality exist. If that definition is adopted, it becomes indispensable to create another word to describe objects, phenomena, or facts that, although not tangible, are present in parallel realities.

The same reasoning applies to the word “real,” as Morpheus's line illustrates: would something be “real” only if perceived by the senses in our reality? Or would it also be real if it were in another parallel reality?

A Bubblelander from a parallel Bubbleland would say: “it is obvious that other Bubblelands do not exist!” And he would be right, if he defined “exist” as being exclusive to whatever is found in his own Bubbleland — bubbles and waves. Thus, other parallel realities, by that definition, would not exist. This would be a Bubblelander like “Aristotle.” A Bubblelander like “Plato,” by contrast, would be thinking of another Bubbleland parallel to his tangible reality.

⁴ Considering that a metaphor is a comparison between two things, aimed at describing one thing in terms of another, we are calling the first thing the “source” and the second thing, the final object of the comparison, the “target.”

1. The semantics of “existing” and the “real”

What is real? How do you define real? If you're talking about what you can feel, what you can smell, what you can taste and see, then real is simply electrical signals interpreted by your brain.

— line spoken by the character Morpheus in the film *The Matrix* (WACHOWSKI; WACHOWSKI, 1999)

Philosophical note. The semantics dealt with here is not merely a linguistic tool. It is an essential component of the theory of knowledge, for understanding meanings is crucial to grasping the limits and possibilities of what can be known. In the philosophy of language, it plays a central role in the discussion of what it means to “exist” and to “be real.”

Semantics is the study of the meaning of words, phrases, and expressions in a given language — a branch of linguistics devoted to the understanding and interpretation of meaning in context.

Moreover, semantics is fundamental to logic, to the philosophy of language, and to artificial intelligence, since it enables the processing and modeling of the meaning of natural language. In sum, it is central to constructing meaningful sentences and to correctly interpreting the world through language.

The Plato of Greek antiquity conceived of the “world of ideas” (PLATO, 2007) as something far more perfect and fundamentally different from tangible reality. This distinction between worlds still echoes today in philosophical discussions about what it means to exist and to be real.

In the image in Figure 2, Plato, with his finger pointing upward, alludes to a reality different from ours: the world of ideas — immaterial, invisible to the senses, and accessible through reason. Aristotle, with his hand held horizontally, represents the empirical, tangible world, accessible through sensory experience. The red and blue pills, alluding to the film *The Matrix* (WACHOWSKI; WACHOWSKI, 1999), reinforce this duality: the red one (the reality pointed to by Plato) suggests an awakening to a transcendent truth, while the blue one (the reality pointed to by Aristotle) symbolizes attachment to the concrete world. This contrast expresses the opposition between Plato’s rational idealism and Aristotle’s empirical realism, in direct dialogue with the *Bubbleland Metaphor*, where multiple realities and natural laws can coexist and interact.

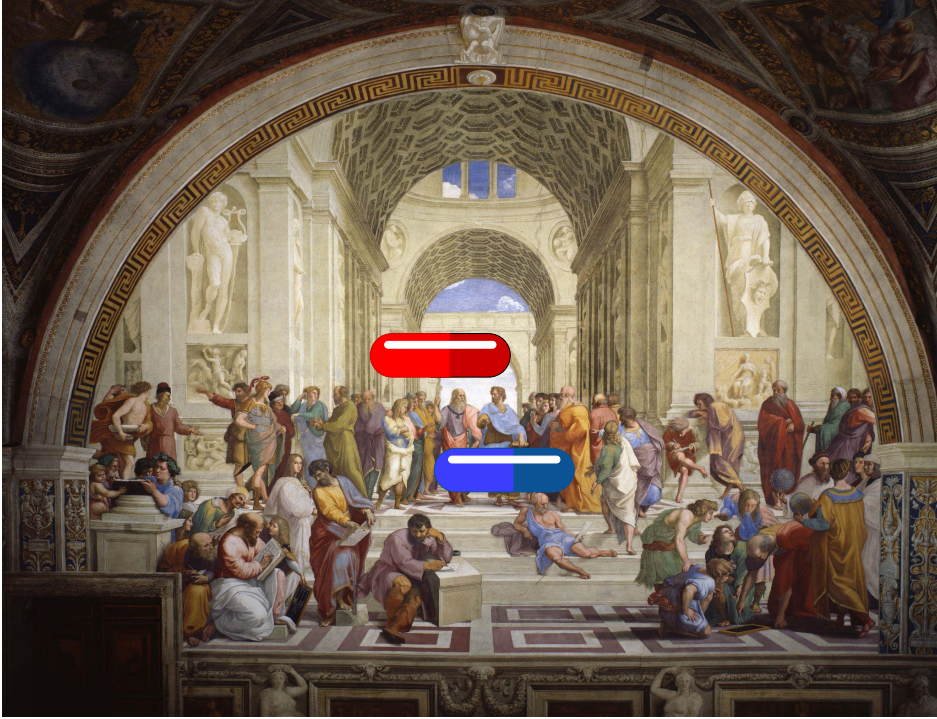


Figure 2 – The image, inspired by Raphael's fresco *The School of Athens* (1509–1511), highlights the contrast between two philosophical realities represented by Plato pointing to the red pill and Aristotle pointing to the blue pill.



Source: Raphael. *The School of Athens*, Stanza della Segnatura, Vatican Museums. Public-domain photograph, via Wikimedia Commons. Image adapted by the author.

1.3.2 Flatland versus Bubbleland

Flatland is a two-dimensional world inhabited by geometric figures, where the shape of each being determines its social standing. Circles represent the highest class, such as priests and leaders, while isosceles triangles, with sharper angles, occupy the base of society, generally as workers or soldiers. Squares and pentagons form the middle class, and the greater the number of sides and the more regular the figure, the higher the individual's social standing. Women are represented by thin straight lines, considered dangerous and inferior — a reflection of the sexism of the era. The work, written by Edwin A. Abbott in the nineteenth century, uses this geometric society as a metaphor to discuss themes such as prejudice, social rigidity, and the human difficulty of comprehending realities beyond our perception — such as the

existence of a third dimension for a being that knows only two (ABBOTT, 1994).⁵

In Bubbleland, also conceived as a two-dimensional universe, the focus is not on social stratification or geometric shapes, but on the cognitive, perceptual, and epistemological limits of its inhabitants. Bubbleland beings are composed of trillions of interconnected bubbles that possess sensory, cognitive, and interactive structures, as well as a mind. Bubbleland posits the existence of multiple two-dimensional surfaces that coexist in a larger space, each governed by its own natural laws, or “nomies”⁶. This is a speculative metaphorical construction that seeks to explore the limits of perception and the challenges of understanding parallel realities.

Table 1 presents a comparison between Flatland and Bubbleland, highlighting their principal conceptual and structural contrasts. Both deal with closed worlds bounded by the perception of their own inhabitants. Whereas Flatland serves as a metaphor for dimensional limitation, Bubbleland proposes a metaphor for the limits of perceived reality and of consciousness, opening the way to multiversal hypotheses.

Table 1 – Comparison between Flatland and Bubbleland

Flatland	Bubbleland	Limitations of the metaphor
A metaphor created by Edwin A. Abbott (1884) to illustrate the limits of perception and to criticize Victorian society. Based on a two-dimensional (2D) universe whose inhabitants are unaware of the existence of the third dimension.	A metaphor inspired by the idea of flatness and multidimensionality, which seeks to illustrate the existence of multiple realities and the limits of our capacity for dimensional perception. The bubbles play a role analogous to the atoms of our 3D world, forming structures on a 2D plane.	Both are conceptual tools, not physical models. Bubbleland does not claim to faithfully reproduce the physics of particles or the real dynamics of soap bubbles.

Continued on next page...

⁵ Published in 1884 under the pseudonym “A Square,” *Flatland* allowed Edwin Abbott to convey an incisive social satire — on class and gender in Victorian England — without compromising his academic and religious reputation. Initially of limited circulation, the work came to be more widely recognized in the early twentieth century, when the spread of relativistic ideas made the notion of dimensions beyond three-dimensional space familiar.

⁶ *Nomy* is a technical neologism introduced in this book to represent the natural laws that govern a specific reality. Its concept will be introduced ontologically in Chapter 6 — The deconstruction of Bubbleland (Page 105), Section 6.3, and in Appendix E (Vol. II) — A mathematical structure for Bubbleland (Page 137).

Table 1 – Comparison between Flatland and Bubbleland (*continued*)

Flatland	Bubbleland	Limitations of the metaphor
The inhabitants are made up of geometric figures, and there is no analogue to a constituent particle.	The inhabitants are “objects made of bubbles,” just as, in human reality, both living beings and inanimate objects are composed of atoms.	Real bubbles are three-dimensional and macroscopic; atoms are quantum and sub-nanometric. The analogy is structural (constituent → whole), not physical.
Each living being is a geometric figure: the number of sides defines the social class.	Each living being is composed of a functional configuration of bubbles, forming objects that interact and organize themselves on the lake.	Real bubbles do not form living beings, nor do they have a metabolism. “Bubbleland life” is an idealization assumed for metaphorical purposes.
The emergence of the third dimension (3D) is inconceivable to the inhabitants, until one of them glimpses it.	The existence of other dimensions or of bubbles external to their local reality is equally inconceivable to the inhabitants of Bubbleland until one of them glimpses it.	Dimensional inconceivability is the most robust point of both metaphors. The limitation lies in the fact that humans <i>already</i> conceive of higher dimensions mathematically, even though they do not perceive them sensorially.
A critique of social rigidity, sexism, and the limits of the scientific and religious thought of the era.	There are no explicit social critiques. Bubbleland’s proposal is a protoscientific production that seeks to explore the limits of molecular modeling, of consciousness, and of computation.	The absence of social critique in Bubbleland is intentional, but it may limit its reach as an instrument of cultural reflection, unlike Flatland.
Perception is limited to what lies within one’s own plane — what is “outside” cannot be seen, only inferred. In Flatland, there is no discussion of parallel realities.	In Bubbleland, the perception of reality is limited to the local Bubbleland, but many events may originate in other Bubblelands, suggesting a new multidimensional epistemology.	The hypothesis of interaction between parallel Bubblelands is speculative and, so far, has no direct experimental support. It is a protoscientific extension.

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Table 1 – Comparison between Flatland and Bubbleland (*continued*)

Flatland	Bubbleland	Limitations of the metaphor
The ignorance of the other dimensions is maintained by social dogmas and fear of the unknown.	The ignorance of external lands or of parallel realities arises from limitations intrinsic to perception and to Bubbleland’s local “physics.”	The metaphor may inadvertently suggest that every episodic limitation is surmountable by a change of perspective. In practice, certain barriers may be structural and permanent (Section 1.3.3.6).

1.3.3 Visual cognition — Flatland versus Bubbleland

There is a subtle difference in visual cognition between Flatland and Bubbleland: in Flatland, a minute perception of the third dimension is needed for vision to be possible⁷. If the beings and other objects in Flatland had no minimal thickness, they could not be visible to one another.

2. Key idea: in Flatland thickness is a contradiction; in Bubbleland, a scale

For one Flatlander to see another, the segment seen must have some thickness — and thickness, in a plane, is the third dimension. But Flatland is a pure geometric idealization: there is nowhere for that thickness to exist. The original metaphor stipulates that its inhabitants see objects of zero thickness, and the stipulation does not hold up.

Bubbleland does not have this problem, and not out of narrative convenience. There the third dimension **does exist**, physically: the bubbles are three-dimensional bodies resting on a surface. What the Bubblelander lacks is **resolution** — the bubbles are far too small, relative to the scale of conscious beings, to be perceived one by one.

There are two layers, and both are necessary. **Scale** explains why the Bubblelander *cannot* perceive the third dimension. **Cognition** explains why he *does not even look for it*: eyes, instruments, and neural circuits were shaped on a flat world, and what they deliver is, for him, everything there is to be perceived.

⁷ In the case of Flatland, everything is metaphorical, but to understand this same difficulty with three-dimensionality, we can imagine the vision of bookworms on the surface of a sheet of paper, as proposed by Gauss. A bookworm would see another bookworm only because of three-dimensionality. This difficulty will be discussed below.

1.3.3.1 The concept of the human cognitive sensory perception system

The cognitive sensory perception system can be understood as the set of biological and mental processes that allow a human being to capture stimuli from the environment, interpret them, and integrate them into consciousness. It involves two fundamental levels:

1. **Primary sensory capture** — carried out by the sense organs (sight, hearing, touch, smell, and taste), which convert physical and chemical stimuli into neural signals.
2. **Higher cognitive processing** — carried out mainly in the brain, where this information is organized, interpreted, compared with prior memories, and placed in a context, generating the conscious experience of reality.

Unlike a mechanical or electronic sensor, the human being not only perceives but also assigns meanings, interpretations, and expectations to the stimuli received. Sensory perception is therefore not a faithful mirror of external reality, but an active and subjective reconstruction of it — shaped by the individual's biology, culture, and life experiences.

1.3.3.2 Human visual cognition

Visual cognition is the process by which the brain interprets and gives meaning to the information captured by the eyes. It goes well beyond the mere reception of light stimuli by the retina: it is a complex operation involving *attention, memory, pattern recognition, inference, and decision-making*.

This process occurs in multiple stages, which include:

1. **Detection of raw visual stimuli** — such as shapes, colors, movements, and contrasts, captured by the retina.
2. **Neural transmission** — the electrical signals generated in the photoreceptors (cones and rods) are sent to the brain via the optic nerve.
3. **Hierarchical processing** — in the visual cortex and associative areas, where the information is organized into objects, faces, three-dimensional spaces, and contexts.
4. **Cognitive integration** — the brain cross-references this information with memories, emotions, and expectations to generate a coherent and meaningful perception of the observed scene.

Visual cognition is highly interpretive and, for that reason, *is not an objective photograph of the world*, but a mental construction that can be influenced by internal

factors (such as attention, beliefs, and emotional states) and external ones (such as lighting, context, or visual ambiguities).

1.3.3.3 Visual cognition in Flatland

In Flatland — a two-dimensional world in which the inhabitants perceive only lengths and angles along a plane — visual cognition would be radically different from ours. This is because its inhabitants would have no direct access to the third dimension (height), which would profoundly limit the way they build mental representations of the world.

A Flatlander's visual perception would be limited to what we might call *linear cognition*: he would perceive objects only as **line segments**. Their lengths and shading patterns (intensity or absence of light) would provide clues about the shape of neighboring objects.

For example, if a square approached a Flatlander observer, he would perceive only the variation of the luminous segment touching his line of sight — like a “bright band” that widens or narrows. The notion of shape or contour could only be deduced *indirectly*, from patterns of variation, memory, and repetition.

Visual cognition in Flatland would therefore involve:

1. **Interpretation of one-dimensional patterns** — the Flatlander brain would learn to recognize “light signatures” characteristic of known objects.
2. **Inference by movement and time** — since there is no visible depth, movement would be essential to deduce the shape and structure of objects.
3. **Mental construction by analogy** — a square would be “seen” as a sequence of segments with abrupt beginnings and endings, interpreted on the basis of prior experience.
4. **Conceptual limits** — a three-dimensional object (such as a sphere passing through Flatland) would be perceived as a sequence of segments that appear out of nowhere, grow, and vanish — something hard to comprehend within the rules of its two-dimensional physics.

Thus, visual cognition in Flatland would be, in essence, an extreme form of *reconstruction of reality by partial inference* — a powerful metaphor for reflecting on our own perceptual limitations with respect to other possible dimensions of reality.

In a purely two-dimensional universe like Flatland, one object's visual perception of another depends exclusively on the tangential intersection between their edges and the line of sight. When a square observes a circle, the only information available

is the linear projection of the visible edge, perceived as a straight segment of apparent thickness d (see Figure 3).

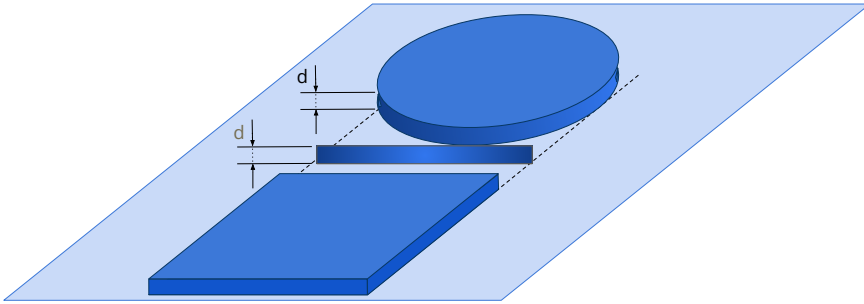


Figure 3 – Projection of a circle as a visible segment for a square: in Flatland, the Square character sees the Circle character as if it were a straight segment of thickness d .



This apparent thickness d tends to zero, as illustrated in Figure 4, since the Flatlander has no cognitive perception of the third dimension. However, d cannot be exactly zero, or the circle would cease to be observed. It follows, then, that there must exist a minimum threshold of perception — geometric or sensory — below which an object ceases to be detectable, even though it continues to exist geometrically. This paradox illustrates the fragility of perception in low-dimensional worlds.



Figure 4 – The thickness d of the visible segment would tend to zero. But if it reached zero, the Flatlander would be unable to see it. The thickness of the segment would not be perceived until the Flatlander conceives of the third dimension.



1.3.3.4 Visual cognition in Bubbleland

In human visual perception (see Figure 5), three-dimensional objects are illuminated by external sources, and part of the light reflected by those objects is captured by the eye's optical system.

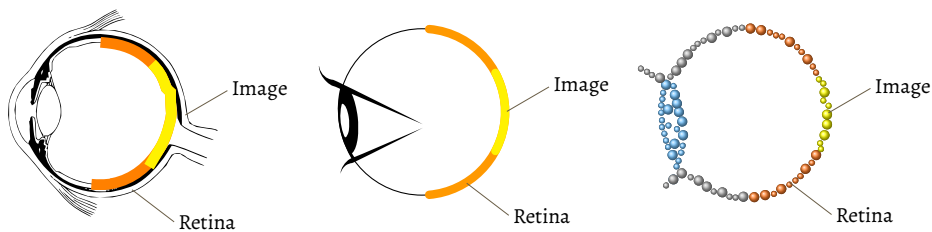


Figure 5 – The “human eye,” a “simplified representation,” and what would metaphorically be a “Bubblelander’s eye.” The bubbles in the third image are greatly exaggerated, merely to make clear that they are the fundamental constituent elements of the Bubblelanders’ eyes.



First image: vectorized from FISCHER, Holly. *Three Main Layers of the Eye*. Open.Michigan, University of Michigan, 2010. Available at: <<https://commons.wikimedia.org/w/index.php?curid=24367145>> . License: CC BY 3.0. Accessed on: 21 Nov. 2025. Retina and image highlights by the author.

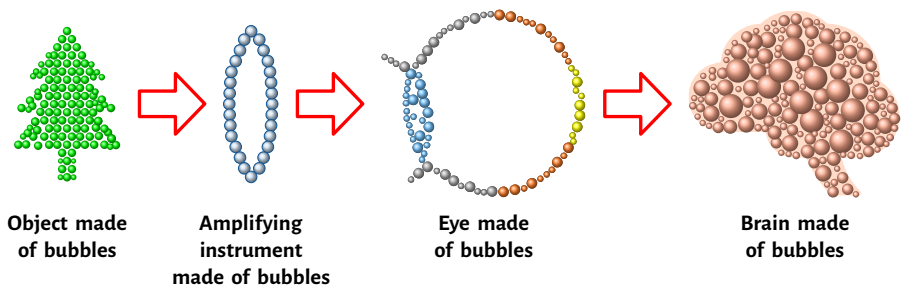


Figure 6 – The Bubblelanders’ eyes are made of bubbles, their brain is made of bubbles, their amplifying instruments are made of bubbles, and everything they can perceive is made of bubbles. But the bubbles are as minuscule to them as atoms are to us.



Light passes through the cornea and the lens, forming an inverted image on the retina. The retina contains photoreceptor cells that convert light stimuli into electrical signals, later interpreted by the brain. What we perceive is not the object itself, but the image projected onto the retina, cognitively reconstructed on the basis of prior experience, depth cues, and sensory integration.

In Bubbleland, all the elements involved in the perceptual process — eyes, optical instruments, brain, and observed objects — are composed of bubbles (see Figure 6). Although the functional structure of the Bubblelanders is entirely formed of bubbles, these fundamental units are extraordinarily small relative to the scale of the conscious beings, just as atoms are to us humans. This means that the Bubblelanders

perceive structures formed by clusters of bubbles, but have no direct cognition of the three-dimensionality of the individual bubbles. Perception results from interactions among sets of bubbles, without conscious access to the internal geometry of each isolated bubble.

Everything in Bubbleland is metaphorical, but our reality is not. When we imagine the cognitive difficulties of the Bubblelanders, we are laying bare our own limitations. The eyes depicted in Figures 5 and 6 do not correspond to functional eyes of a real Bubbleland; metaphors are not factual descriptions, but tools of analogy. The human analogue, however, is real: we face genuine difficulties in seeing the absurdly small — bubbles for the Bubblelanders and atoms for us — and also difficulties in comprehending additional dimensions. This is one of the central aims of the metaphor.

From a visual standpoint, the Bubblelanders see objects as line segments. But unlike the Flatlanders (who, ideally, would perceive nothing, since their objects would have zero thickness, as discussed in Section 1.3.3.3), the Bubblelanders perceive these segments as all there is to be perceived, because their cognitive faculties, their ocular apparatus, and their neural circuitry were shaped that way.

Analogously, we humans have no cognitive perception capable of representing a fourth spatial dimension. Everything we see seems to us to be all that can be seen — and, for the human ocular apparatus and for our neural learning, it is exactly that.

1.3.3.5 “Human instruments” versus “Bubbleland instruments”

Table 2 presents some of the instruments the human species has developed to investigate physical reality.

Table 2 – Description of the main instruments and the type of image generated⁸

Instrument	Brief description	Type of image generated
Optical microscopy	Uses visible light and glass lenses to magnify samples.	2D images and, in some techniques, shallow 3D.
Electron microscopy	Uses electron beams, allowing nanometric resolution.	High-resolution 2D images.
Scanning tunneling microscopy (STM)	Measures the tunneling current between a conducting tip and the surface of conducting samples.	2D topographic maps of the surface (local density of states).
Nuclear Magnetic Resonance (NMR/MRI)	Detects the magnetic response of nuclei in an intense field.	3D volumes (tomography).
Particle Accelerators	Accelerate particles and record collision events.	3D reconstructions of trajectories.
X-ray Crystallography	Records diffraction patterns for structural reconstruction.	3D maps of electron density.
Mass Spectrometry	Separates molecules by mass/charge.	2D spectra; some methods generate 2D chemical images.
Other methods	Various related instrumental techniques.	Limited to images up to 3D.

All these devices — optical and electron microscopes, spectrometers, particle accelerators, among others — are designed to detect and represent objects composed of atoms or subatomic particles, invisible to the naked eye. A more detailed description of these and other techniques — including their physical principles, typical resolutions, and limitations — is presented in Appendix AL (Vol. III) (Techniques for the structural determination of biomolecules, Page 427), with a specific focus on the structural determination of biomolecules. The comparative table available there (Table 36, Vol. III, p. 433) makes it possible to see how each human instrument operates within the same dimensional limits discussed in this section: all produce, at most, 2D representations or 3D reconstructions.

Therefore, even with significant technological advances, **none of these instruments would be capable of directly detecting objects situated in a fourth spatial**

⁸ **Technical note:** Regardless of the physical principle used (optical, electronic, quantum, magnetic, or radiative), the final product of these instruments is always limited to 2D or 3D representations. No instrumental technique is capable of producing 4D spatial structural images.

dimension. Analogously, in Bubbleland, any instrument built out of bubbles would be restricted to perceiving two-dimensional structures composed of bubbles. Thus, the three-dimensional human reality would remain completely inaccessible to a Bubblelander, even if he had highly advanced observation tools made exclusively of bubbles.

This finding reveals a fundamental limitation of Flatland as a metaphor: it allows us to reflect on dimensional perception, but it is not able to explore the limits of its inhabitants' own cognitive and instrumental apparatus. Bubbleland, on the other hand, offers a more powerful metaphorical structure, for it allows us to generalize the concept of perceptual limitation beyond geometry, encompassing also the nature of the observation instruments and the cognitive limits imposed by the observer's material constitution. This is, therefore, one of the conclusions that could not have been derived from Flatland, and that make evident Bubbleland's expanded explanatory capacity.

1.3.3.6 Cognitive limits as epistemic limits

The *Bubbleland Metaphor* is not merely an image for thinking about dimensions and hidden structures. It is also a tool for making explicit a more general thesis: **cognitive limits impose epistemic limits**.

In strict terms, a *cognitive limit* is any restriction intrinsic to the observer — perceptual, neural, conceptual, linguistic, or imaginative — that defines *how* the world can be represented internally. An *epistemic limit*, in turn, is a restriction on *what* can be known, justified, or decided under criteria of rationality, evidence, and coherence. The thesis is that the latter derives, in large measure, from the former: *no one can know (in the strong sense) that which they cannot even represent in a minimally stable way*.

3. Key idea: Cognitive limits impose epistemic limits

The science of any observer confined to a particular reality is, necessarily, an **interface science**: it describes what manifests, not what sustains the manifestation.

This limitation is not technological — it is **structural**. It is not a matter of building better instruments, but of recognizing that the observer's own cognitive apparatus defines the horizon of the conceivable. What cannot be internally represented cannot be known in the strong sense, even if it exists and produces observable effects.

continued →

3. Key idea: Cognitive limits impose epistemic limits (*continued*)

In Bubbleland: the Bubbleland physicist formulates interactions between bubbles with high precision, but has no access to the air, the water, and the surfactant that sustain the bubbles (Section 1.4).

In our reality: QED predicts the magnetic moment of the electron to fourteen decimal places, but does not tell us *what* the electron is (Subsection 1.4.2).

This concept is taken up and formalized in Chapters 6 (nomies), 10 (expanded epistemological classification), 11 (ontological limits), and 18 (general synthesis), besides figuring among the irreducible pillars of the work (Appendix AH, Vol. III).

Bubbleland makes this visible. The Bubblelander suffers not merely from a lack of instruments: he suffers from a lack of adequate *categories*. Even if an external phenomenon produces consistent internal effects, it will tend to be interpreted through a local conceptual repertoire, built up over an evolution and a culture shaped for a planar world. The result is a confined epistemology: a universe of effective explanations that are nonetheless ontologically partial.

The same logic applies to the human. Even when we have instruments that extend perception — microscopes, accelerators, tomographies, spectra — the final product remains a representation compatible with our cognitive schemas: 2D images, 3D reconstructions, curves, time series, probabilities, and models. Instrumentation broadens the *interface* with the real, but does not eliminate the final bottleneck: *human interpretation*. The limitation lies not only in the reach of the measurement, but in the way the brain manages to organize what has been measured into intelligible concepts.

This explains why certain frontiers of knowledge are not merely technological, but structural. There are at least four ways in which cognitive limits turn into epistemic limits:

1. **Limit of conceivability:** without a conceptual space for a hypothesis, it does not even enter the domain of the thinkable. Ideas can be true and yet remain outside the cognitive horizon of a species or a culture.
2. **Limit of expressibility:** even when something is intuited, there may be no language, mathematics, or metaphor capable of encoding it with sufficient precision for public discussion and intersubjective validation.
3. **Limit of instrumental interpretability:** data may exist but be indistinguish-

able from noise, artifact, coincidence, or ambiguity. The problem is not the absence of signal, but the absence of a *model* that makes it legible.

4. **Limit of explanatory closure:** a theory may predict with extreme precision and yet offer no internal ontology. It closes the circuit of the *how* without touching the *what it is*. This kind of closure may be the maximum attainable under certain cognitive constraints.

4. Perception of the fourth dimension and the limits of Flatland

If atoms or biomolecules were intrinsically four-dimensional, human beings still would not perceive that fourth dimension directly. The human visual system evolved to interpret only the macroscopic effects of immense clusters of particles — atoms and molecules — which, taken together, organize into stable three-dimensional structures. Even if advanced instruments produced 3D projections of 4D objects (with temporal and geometric distortions revealing that additional dimensionality), human cognition would face profound difficulties in interpreting them: our entire sensory-cognitive apparatus was shaped by Evolution and by neural learning to recognize solidly three-dimensional patterns, not hypervolumetric ones.

Something analogous happens in Bubbleland. Bubblelanders see only the macroscopic patterns produced by clusters of three-dimensional bubbles which, taken together, form an essentially two-dimensional surface. Their cognitive systems, like ours, were shaped to interpret the world in its macroscopic dimensionality — in their case, 2D. Even if instruments made of bubbles were capable of producing two-dimensional projections of 3D structures (with specific distortions that would betray that additional depth), the cognitive interpretation would likewise be limited by a perception trained for a planar world.

It is precisely here that Flatland stands apart: in the original metaphor, its inhabitants are not made up of smaller particles, nor do they perceive clusters of constituents. They would simply *have to see* segments and polygons of zero thickness, because the very geometric construction of Flatland defines them that way — which would be a contradiction, since nothing of zero thickness can be seen (Box 2). In that universe there is no hierarchy of scales giving rise to aggregate perception and its limits. Therefore, discussions about 4D particles, tunneling, or perceptual projections of higher dimensions make sense only in universes like ours and the Bubblelanders', where perception is the product of cognitive systems evolved over invisible microscopic layers, and not in a geometric idealization like Flatland.

From this standpoint, *Bubbleland* is a metaphor about science as interface science: we describe with rigor what the world allows us to observe and manipulate, but

this does not guarantee access to the ultimate structure. The transition between *description* and *ontology* is exactly the point where cognitive limits tend to manifest as epistemic limits.

This conclusion does not diminish the value of science; on the contrary, it delimits its reach honestly. Science remains the best method known for filtering out illusions, controlling biases, and producing robust predictions. Yet the metaphor suggests that *not every limit is a failure*: some limits may be an unavoidable property of any intelligence situated in a particular reality, with finite senses, finite language, and neural mechanisms shaped by local pressures.

Thus, the *Bubbleland Metaphor* plays an additional role: it not only illustrates hidden dimensions, but also makes explicit that the map of possible knowledge is drawn, in part, by the very organism that attempts to know.

5. Time and Emergent Normality

A recurring principle across different levels of reality is that the macroscopic whole may appear entirely normal, even while being sustained by microscopic entities of a deeply exotic nature. Just as billions of atoms form familiar three-dimensional objects, and just as countless three-dimensional bubbles produce, for the Bubblelanders, a stable two-dimensional world, so too time may conceal, in its most intimate structure, characteristics very different from those we perceive.

If the microstructure of time were granular, discontinuous, asymmetric, or endowed with properties bearing no resemblance to the smooth, unidirectional flow we experience, this would not become evident to us. Evolution shaped our senses and our cognition to interpret only the macroscopic behavior of time — an emergent average of processes that could, at their minute scale, be radically different. What we call an “instant” may be as far from the fundamental structure of time as the solid appearance of a table is from the quantum dynamics of its constituents.

This perspective sheds light on the classic paradox raised by Aristotle: if time is composed of parts that do not persist — a past that no longer is, a future that is not yet, and a present that vanishes when divided — how can it exist? Perhaps the answer lies in the very way we perceive: what seems to us continuous and simple may be merely the emergent effect of a microscopic reality whose deep nature remains invisible.

1.3.3.7 The cognitive perception of time

The reflection in Box 5 allows us to reexamine one of the oldest questions about time.

The tension between perceived continuity and the apparent nonexistence of its

parts makes the Aristotelian paradox especially intriguing in light of the idea that what we experience may be merely the emergent aspect of a deeper, inaccessible temporal structure.

1.4 External equations and hidden ontology

A Bubbleland physicist, when studying the bubble and trying to write an “equation for the bubble” from his own point of view, can describe the interaction forces among the bubbles, but he would not be able to describe the bubbles themselves, since he is unable to imagine what the water and the surfactant that allow the bubbles to exist might be, much less the air that must necessarily exist inside them. Let us call all of this that sustains the bubble’s existence its “internal structure” — it is invisible and inaccessible from within, only from without.⁹

This physicist would be confined to his own 2D reality — the surface of Bubbleland — and could therefore only model the behavior of the bubbles based on the interactions observable within that plane. He would perceive tensions, deformations, attractions, and repulsions — and might even formulate effective laws to describe such phenomena. Those laws, however, would always be approximations of something more fundamental. He would have access neither to the idea of the third spatial dimension, where the air accumulates and gives the bubbles their volume, nor to the internal structure of those bubbles — elements entirely transcendent to his perception.

From an epistemological standpoint, his physics would be phenomenological, not ontological — he would describe what he sees, but could not explain why it exists or what its ultimate foundations are.

This analogy serves well to illustrate the cognitive limitation of observers within a projected reality. Just as this Bubbleland physicist cannot conceive of the dimension where the internal structure of his bubbles resides, we humans too, in modeling our particles and fields, may be dealing with projections or manifestations of something that escapes our perceptible dimensions.

When a “Bubbleland Paul Dirac” writes a “bubble equation,” he is describing fields associated with the bubble, that is, the influence it exerts on the surface of Bubbleland — perhaps something analogous to a 2D scalar or vector field, representing

⁹ A much more detailed discussion of this subject will be given in Chapters 11 and 12. On the “internal structure” discussed here, see Figure 48, in Section 12.5.3. Figure 46, in Chapter 11, is also relevant for highlighting the analogy “Bubble” $\longleftrightarrow$ “Fundamental particle.”

presence, charge, deformation, or another locally observable property. That equation, however, would not describe the essence of the bubble — only the observable effects it produces. The internal structure — that which gives rise to its properties — would remain completely hidden from his model.

Even the deepest equations (such as a Bubbleland version of the Dirac equation, which in our reality already touches the boundary between the classical and the quantum) would be merely projected shadows of a more complex reality that transcends the surface of the bubble.

Thus, the “bubble equation,” however elegant, would be no more than a partial description — perhaps even sufficient for internal predictions, but incapable of reaching the true ontology of the bubble. It is like trying to explain electricity by observing only the sparks — without knowing that electrons exist.

Beyond the epistemological difficulty of conceiving the real structure of the bubbles, there is also a difficulty of computation, equally profound.

Even if the Bubbleland physicist were extremely advanced and attempted, by means of differential equations, to simulate the behavior of the bubbles based solely on the effects observed at the surface, he would run into an almost insurmountable complexity¹⁰:

1. **Highly stiff equations (stiff systems):** small variations in certain variables would demand enormous variations in others to keep the solution stable. This imposes minuscule integration steps in numerical simulations, making the computation time unfeasible even with massive parallelization.
2. **Very high dimensionality:** in trying to represent the collective behavior of bubbles, or even the internal behavior of a single bubble, he would be dealing with thousands or millions of coupled variables, perhaps with degrees of freedom that do not even belong to the observable two-dimensional space. The computational cost grows exponentially (the curse of dimensionality).
3. **Impossibility of analytical solutions:** unlike equations such as Schrödinger’s or Dirac’s, which have exact solutions only in extremely simple cases, here not even viable simplified models would exist. Any attempt at a simplifying equation would entail a loss of fidelity.
4. **Dependence on unobservable parameters:** critical parameters — such as the properties of the internal air, the viscosity of the film, the temperature, and so

¹⁰ Some of the problems mentioned here are detailed in Appendices V (Vol. II) (Types of “stubborn problems” in numerical computation, Page 387) and W (p. 407) (Numerical methods for ODEs/PDEs: a practical guide).

on — would be entirely beyond sensory and experimental reach. The equation might even exist, but it would be like trying to solve $f(x, y, z, w, \alpha, \beta, \gamma) = 0$ without any idea of the value of α, β, γ .

5. **Emergent, non-reducible behaviors:** even if he solved part of the equation for an isolated bubble, he would not be able to extrapolate it to multiple bubbles. Nonlinearity and coupling effects would give rise to collective phenomena that do not decompose into individual solutions.

In other words: even with physical intuition, even with approximate equations, and even with advanced computational technology at his disposal, he would still be locked inside a mathematical labyrinth in which the complete solution would remain, in practice, inaccessible. We will, however, return to this subject in Chapter 11, where we will discuss alternatives for solving this problem.

1.4.1 The communicative role of the metaphor

The *Bubbleland Metaphor* serves our reality with surgical precision: perhaps many of the fundamental laws of nature lie beyond our ability to formulate — and, even if formulated, beyond our ability to solve. Explaining all of this to other people without resorting to the metaphor is absurdly difficult. Without it, in trying to address the limits of current science, of human epistemology, or of the computability of complex systems, we inevitably enter into concepts that demand an uncommon technical familiarity and philosophical openness.

1.4.2 The ontology of the electron: a provocation to be developed later

The *Bubbleland Metaphor*, when applied to the electron, suggests considerably bold ontological implications. We know how to formulate the electron in “external” terms — its fields, its interactions, its wave function, its spin, its magnetic moment. The Dirac equation is one of the most beautiful achievements of physics, and Quantum Electrodynamics (QED) predicts the magnetic moment of the electron to fourteen decimal places of precision. But all of this concerns how the electron manifests to the observer, not what it is in essence.

We model the outline, but we are ignorant of the content. This is exactly what happens with the Bubbleland physicist: he measures forces, tensions, and deformations, but does not access what gives rise to those properties. In questioning this point, one is not denying QED or its remarkable predictive power. What is placed

under discussion is only the *ontological* level of the description: current science is, to this day, a science of the interface, not of the essence.

This observation is deliberately left in suspense in this chapter. Its systematic treatment requires the framework of the layered structure and the quantum reinterpretation, which will be developed in Chapter 11 (The composition of our reality) and, above all, in Chapter 12 (Layered structure and quantum reinterpretation). The reader interested in the more radical ontological reinterpretation may consult directly Section 12.12.1 of that chapter; the others will find there the complete path, with the appropriate epistemological disclaimers.

1.4.3 Bubbleland: an image for thinking the unthinkable

When we say: “imagine a 2D physicist trying to write an equation for the bubble, without knowing what air, water, or surface tension are...,” a concrete and intuitive image lights up. The person understands, by analogy, the limitation of the internal observer. They understand that the bubble makes sense from the standpoint of one who is outside — but for one who is inside, all that remains is the projected shadow of reality.

This metaphor offers a cognitive bridge and an accessible language for addressing the limits of fundamental physics, computational difficulties, and even philosophical and theological questions¹¹. It is flexible and expandable — a mental laboratory for thinking the unthinkable.

¹¹ Although certain themes may coincide with traditional questions of philosophy and religion, this work does not propose spiritual interpretations, but physical hypotheses still under development.